Matteo Bonotto, Ph.D.

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About

I am a **Senior AI Research Engineer** specialized in large language models, deep learning systems, and high-performance scientific computing. My work spans the full LLM stack, from designing and implementing custom Transformer architectures and training methods to GPU kernel optimization, distributed training, evaluation, large-scale inference, and agentic harness design.

At Giotto.ai, I work on LLM architecture and reasoning research, distributed training on multi-node GPU clusters, and high-performance inference systems. I have implemented Transformer models and training components from scratch, developed custom Triton kernels achieving up to 10× speedups and 18× memory-efficiency improvements. I worked extensively with distributed training and inference of >1T parameter models.

Prior to my work on LLMs, I worked in industrial AI and magnetic-confinement fusion research. My research included machine-learning methods for tokamak disruption prediction and the development of FEM-based numerical tools for plasma equilibrium, electromagnetics, and plasma control. This background gives me a strong foundation at the intersection of machine learning, numerical methods, scientific computing, and high-performance computing.

Skills

LLM & Deep Learning Research: Transformer architectures, LLM pre-training and fine-tuning, reasoning systems, Physics-Informed ML

Distributed Training & HPC: PyTorch Distributed, DDP, FSDP, Megatron-LM, tensor/pipeline/data parallelism, Ray, torchrun, multi-node GPU training

LLM Inference & Optimization: vLLM, Ray Serve, FP8 and low-bit quantization, KV-cache optimization, speculative decoding, continuous batching, long-context inference, model parallelism, inference benchmarking

GPU & Performance Engineering: Triton, GPU kernel optimization, memory and compute profiling

Scientific Computing: Finite Element Methods, numerical methods, electromagnetics, plasma equilibrium and control, high-performance computing

Programming: Python - expert; MATLAB/Simulink - expert; Fortran

Frameworks & Tools: PyTorch, Hugging Face Transformers, TRL, PEFT, TensorFlow, Docker, Git, MLflow, Airflow, GCP, Linux

Software Engineering: Research software architecture, profiling and performance optimization, containerized ML systems, CI/CD, reproducible training and inference pipelines

Education

University of Padua — Padua, Italy
Ph.D in Magnetic Confinement Fusion; — *Sep 2016 – Mar. 2020*

Ph.D. 2nd best thesis prize: Thesis *Modeling of MHD instabilities in existing and future fusion devices in view of control*, at Hic-Labor prize (Vicenza Olympic Academy, 2021) .

University of Padua — Padua, Italy
M. Sc. in Electrical Engineering; ***GPA: 3.91/4.00*** — *Sep 2012 – Mar. 2015*

University of Padua — Padua, Italy
B. Sc. in Energy Engineering; ***GPA: 3.51/4.00*** — *Sep. 2009 – Sep. 2012*

Experience

Giotto.ai — Lausanne, Switzerland
Senior AI Research Scientist — *Apr. 2024 – Present*

- Core contributor to in-house LLM training and post-training pipeline. Multi-GPU and multi-node cluster training using FSDP, Megatron-LM, Ray, torchrun, and tensor/pipeline/data parallelism.

- Core contributor to Giotto-OS serving infrastructure: developed scalable LLM inference infrastructure, including tensor parallelism, FP8 quantization, speculative decoding, prefix caching, chunked prefill, KV-cache optimization, and long-context serving. Deployed models up to >1T parameters.
- Core contributor to Giotto-OS agentic harness.
- Performed research and work on Transformers and implemented custom architectures. Contributed to solutions awarded gold medals in the Kaggle ARC Prize in 2024 and 2025.
- Develop and optimize GPU kernels with Triton, achieving up to 10× speedups and 18× improvements in memory efficiency for critical model operations.
- Work on model compression and deployment, including FP8 and low-bit quantization, memory-efficient model loading, and optimization of models for production inference
- Develop evaluation and experimentation pipelines for reasoning and multimodal models, including ARC-style abstract reasoning, mathematical reasoning, structured-image understanding, and custom continuous/agentic evaluation harnesses
- Lead and mentor a team of 4 researchers, coordinating research directions, experimental work, and implementation

Eni Milan, Italy

AI Scientist *Jan. 2023 – Mar. 2024*

- Designed and implemented custom AI solutions for R&D and industrial applications. Collaborated with MIT scientists to develop a tokamak disruption prediction algorithm that outperformed state-of-the-art approaches by **5%** in accuracy.
- Mentored junior teammates and shared ML best practices in a team of 30.

National Institute for Nuclear Physics Padua, Italy

Research Scientist *Jan. 2021 – Dec. 2022*

- Mathematical modeling for plasma physics, electromagnetics, and magnetic confinement fusion.
- **Developed and open-sourced** a physics-informed-ML-based model for plasma equilibrium achieving 10x speedup compared to SOTA.

RFX Consortium Padua, Italy

Research Scientist *Oct. 2019 – Dec. 2020*

- **Developed and open-sourced** a FEM-based tool for plasma equilibrium with novel formulation for boundary conditions.

General Atomics San Diego

Visiting Ph.D. Student *Sep. 2019 – Dec. 2019 and May 2020 – Aug. 2020 Dec. 2020*

- Advanced mathematical models for coupling MHD and electromagnetic phenomena. **Expanded modeling capabilities** of existing linear MHD code to model non-linear growth rates of Resistive Wall Modes plasma instrabilities.

Dep. of Information Engineering - Univ. of Padua Padua, Italy

Research Scientist *Jun. 2015 – May 2016*

- Model order reduction to large-scale electromagnetic simulations for control applications. **Expanded modeling capabilities** of existing static plasma state estimation tool to estimate plasma states dynamically along the entire plasma discharge.

Awards & Achievements

Ph.D. thesis prize: 2nd place for best Ph.D thesis at 2021 Hic-Labor by the Vicenza Olympic Academy. (Jul 2021)

High Honors Degree: Awarded to M.Sc. alumni who have graduated with a GPA $\geq$ **3.73/4.00** and a final graduation $>$ **113/110**. (Mar. 2015)

Outstanding M.Sc. GPA: Completed the M.Sc. degree with final GPA of **3.91/4.00**, among the highest of the cohort.